

Database Learning Notes

SQL, MySQL and NoSQL | Banking examples | Prepared for Kush

1. What is a database?

A database is an organized collection of data. A banking application can store customers, accounts and transactions in a database. A database management system (DBMS) provides tools to store, retrieve and manage that data.

2. Why use a database?

A database makes it easier to insert, search, update and delete records. It also supports persistent storage, rules for valid data, and controlled access. Data saved in a database can remain available after the application closes.

3. What is a relational database?

A relational database organizes data into tables with rows and columns. Tables can be related using keys. For example, an accounts table can contain a `customer_id` that refers to a customer in a customers table.

Term	Meaning	Banking example
Database	Collection of related tables	banking_db
Table	Rows and columns describing an entity	accounts
Row	One record	One account
Column	One attribute	balance
Primary key	Uniquely identifies a row	account_id

4. SQL versus MySQL

SQL stands for Structured Query Language. It is used to query and manage relational databases. **MySQL** is a relational database management system that understands SQL. Oracle Database and PostgreSQL also use SQL, although some syntax differs between systems.

Remember: SQL is the language. MySQL is the database system.

MySQL: Our Banking Example

5. Create a database and a table

```
CREATE DATABASE banking_db;

USE banking_db;

CREATE TABLE accounts (
    account_id INT AUTO_INCREMENT PRIMARY KEY,
    customer_name VARCHAR(100) NOT NULL,
    balance DECIMAL(12, 2) NOT NULL DEFAULT 0.00
);
```

SQL term	Explanation
INT	Stores an integer.
AUTO_INCREMENT	Automatically generates the next ID when it is omitted.
PRIMARY KEY	Requires a unique, non-null identifier for each row.
VARCHAR(100)	Stores variable-length text up to 100 characters.
NOT NULL	Does not allow a NULL (missing) value.
DECIMAL(12, 2)	Stores exact numbers: 12 total digits, including 2 after the decimal point. Suitable for monetary amounts.
DEFAULT 0.00	Uses 0.00 when the balance is omitted from an insert.

6. Insert and read an account

```
INSERT INTO accounts (customer_name, balance)
VALUES ('Kush', 1500.00);
```

```
SELECT * FROM accounts;
```

INSERT adds a row. SELECT reads data. The asterisk (*) requests all columns. For a new empty table, the first generated account_id is normally 1.

account_id	customer_name	balance
1	Kush	1500.00

CRUD and NoSQL

7. CRUD: the four basic operations

CRUD means Create, Read, Update and Delete. Here, Create means adding a record with INSERT; CREATE TABLE defines the table itself.

```
-- Create a record
INSERT INTO accounts (customer_name, balance)
VALUES ('Aman', 500.00);

-- Read one account
SELECT * FROM accounts WHERE account_id = 1;

-- Update one account
UPDATE accounts
SET balance = 1700.00
WHERE account_id = 1;

-- Delete one account
DELETE FROM accounts WHERE account_id = 2;
```

Important: WHERE chooses which rows to update or delete. Without WHERE, these UPDATE or DELETE statements affect every row in the table. The update above directly sets a balance; a real banking application needs additional validation and transaction handling.

8. What is NoSQL?

NoSQL is commonly explained as "Not Only SQL." NoSQL databases use models such as documents, key-value pairs, graphs or wide columns. They also support CRUD operations. Their data organization and query methods differ from relational databases.

MongoDB stores documents that look similar to JSON:

```
{
  "accountNumber": "ACC101",
  "customerName": "Kush",
  "balance": 1500,
  "accountType": "Savings"
}
```

Relational databases	NoSQL databases
Tables with rows and columns	Documents, key-value pairs, graphs or wide columns
Usually have a predefined table schema	Often allow a flexible schema; validation is still possible
MySQL, Oracle Database, PostgreSQL	MongoDB, Redis, Cassandra

Quick revision: Database = organized data. SQL = language. MySQL = relational database system. MongoDB = document database. CRUD = insert, read, update and delete.